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Paleomagnetic results from Cretaceous and Jurassic rocks of South and Southwest Yunnan: evidence for large clockwise rotations in the Indochina and Shan–Thai–Malay terranes

Kainian Huang [1] and Neil D. Opdyke

Department of Geology, University of Florida, Gainesville, FL 32611, USA

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ABSTRACT

Cretaceous and Jurassic redbeds have been sampled from South and Southwest Yunnan, China, for paleomagnetic study. The paleomagnetic directions resolved from the Cretaceous rocks indicate that the Xiaguan area, which is located on the Red River fault, has not been rotated with respect to stable Eurasia since at least Late Cretaceous times and that the Jinggu–Mengla area 200–400 km farther south has been rotated by 46–65° clockwise. The Jurassic results reveal the same rotation pattern. At face value, the differences between the observed Cretaceous paleolatitudes and those predicted from the Eurasian apparent polar wander path (APWP) indicate a southward extrusion of the region, but the amount of extrusion is not large enough to be resolved by paleomagnetic methods. Comparison with Mesozoic paleomagnetic data from central Yunnan, northern Thailand and northwestern Borneo suggests that the Red River fault is not a demarcation between unrotated and significantly rotated regions, and that Sundaland did not respond to the India–Asia collision as a single coherent unit. The existing paleomagnetic data from East Asia appear to indicate that the crust in central Tibet and Tien Shan north of Tarim directly in front of the Indian indentor has been substantially shortened and thickened whereas the crust in the syntaxial areas has probably been largely squeezed out. To further evaluate the relative importance of crustal thickening and extrusion in the tectonic evolution of the Tibetan plateau, paleomagnetic investigation should be intensified in the bounding zones to the Indian indentor and be expanded into the Cenozoic sedimentary rocks in Tibet and surrounding regions.

1. Introduction

The early Tertiary collision of the Indian subcontinent with Eurasia has a profound impact on the Cenozoic tectonics of Asia [1–3]. One of the major predicted consequences is large clockwise rotation and southeastward extrusion of Sundaland, which includes the Indochina and Shan–Thai–Malay terranes [2,3]. Achache et al. [4] were the first to test this prediction using paleomagnetic data available at that time and concluded that the large clockwise rotation of Indochina was supported. A later study of the rock formations with ages spanning the Devonian to Jurassic from the Khorat plateau of Thailand, however, revealed pervasive remagnetization and thus discredited the previously published results from the Triassic to Cretaceous [5]. In the meantime, McCabe et al. [6] conducted a paleomagnetic investigation on the late Neogene basalt from Thailand that showed differential rotation between western and central Thailand versus the Khorat plateau, whereas Schmidtke et als'. study [7] on the Upper Jurassic to Miocene rocks from northwestern Borneo revealed counterclockwise rotations of more than 100°. Apparently, paleomagnetic data from this region is not yet robust enough to confirm the large clockwise rotation of this region.

We present in this paper Cretaceous and Jurassic paleomagnetic results from southern and southwestern Yunnan province that demonstrate

[1] Permanent address: Institute of Geology, Academia Sinica, Beijing 100029, China

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unequivocal large clockwise rotations, and place them in a regional framework in an attempt to test differing models regarding the effects of the India–Asia collision on the tectonics of East Asia.

2. Geology and sampling

The NW–SE trending Red River fault in Yunnan province marks the southwestern boundary of the South China Block (SCB) (Fig. 1). South and Southwest Yunnan to the southwest of the fault zone comprise two tectonic units, the Simao Depression and the Baoshan Block, the former being part of the Indochina Terrane and the latter the northern extension of the Shan–Thai–Malay Terrane [8]. Within China, the Baoshan Block is bounded in the west by the Nujiang Suture and in the east by the Changning–Shuangjiang Suture [Zhong Dalai, pers. commun., 1992]. The Precambrian basement is not exposed,

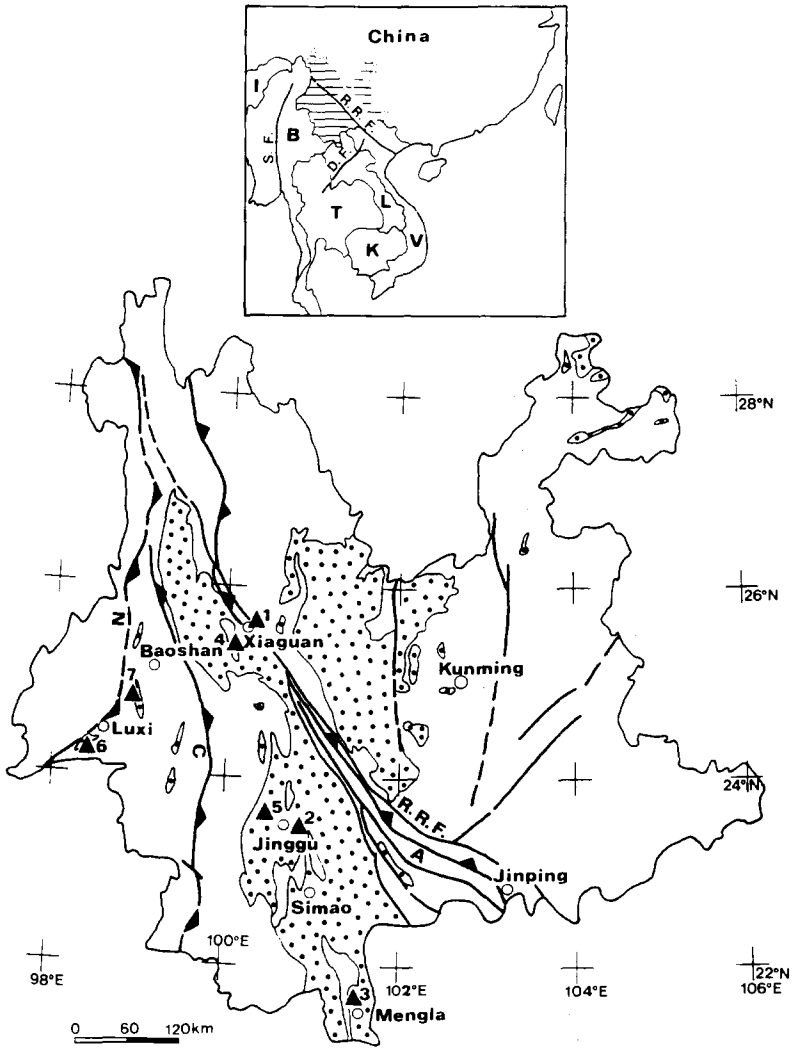


Fig. 1. Schematic geological map of Yunnan province. Numbered triangles denote the sampled sections of this study: *1* = Lake Erhai; *2* = Yangdiaoyu; *3* = Longlin; *4* = Shuilazhuqing; *5* = Wenzhaoying; *6* = Mengga–Longhai; *7* = Datianyakou. Heavy lines are major faults. Heavy lines with black teeth are proposed sutures: *A* = Ailaoshan; *C* = Changning–Shuangjiang; *N* = Nujiang. Stippled area indicates Jurassic–Cretaceous outcrops. The inset shows the regional map of Southeast Asia: *B* = Burma; *I* = India; *K* = Kampuchea; *L* = Laos; *T* = Thailand; *V* = Vietnam. Three major faults are also indicated: *D.F.* = Dien Bien Phu fault; *R.R.F.* = Red River fault; *S.F.* = Sagaing fault. Yunnan province is shaded.

and the cover strata are characterized by folded Cambrian to Triassic platform carbonates and clastics with subordinate volcanics [9]. The Jurassic is of limited areal distribution, and the Cretaceous and Paleogene are missing. The lignite-bearing Neogene unconformably overlies the Jurassic and older rocks [10].

In contrast, the metamorphosed Paleozoic rocks in the Simao Depression outcrop as a narrow zone along the Ailaoshan Suture which runs parallel to the Red River fault [11]. Exposures of Carboniferous to Triassic clastic and carbonate rocks of the marine facies occur sporadically. Jurassic through to lower Tertiary red continental sedimentary rocks dominate the stratigraphic succession. This thick Jurassic through lower Tertiary sequence is known as the West Yunnan Redbeds [8]. The molasse deposits of Eocene–Pliocene age unconformably overlie the redbed sequence.

In the Simao Depression, the total thickness of the Jurassic–Cretaceous sequence is over 6000 m [12]. The Jurassic succession is composed of three formations, from bottom to top the Lower Jurassic Yangjiang Formation, the Middle Jurassic Hepingxiang (in the western part of the depression) or Huakaizuo (in the eastern part) Formation, and the Upper Jurassic Bazhulu Formation [13,14]. The overlying Cretaceous sequence is also divided into three parts [15]: the Lower Cretaceous is designated as the Jingxing Formation, the middle Cretaceous includes the Mangang and Pashahe formations, and the Upper Cretaceous is represented by the Mankuanhe Formation. The age of these rocks is primarily controlled by such fauna as bivalves, gastropods and ostrapods [14–16]. The Middle Jurassic Hepingxiang/Huakaizuo Formation and the Lower Cretaceous Jingxing Formation are especially useful in correlating and mapping this thick redbed sequence due to their variegated color and rich fossil content.

The Jurassic stratigraphy on the Baoshan Block, on the other hand, is of a marine facies [10]. The Lower Jurassic is absent. The Middle Jurassic includes three formations, which are, from bottom to top, the Mengga, Liuwan and Longhai formations. The Mengga Formation is 545–1000 m thick and consists mainly of purplish red sandstone and siltstone. The overlying Liuwan Formation is made up of limestone, and contains abundant brachiopods such as *Holcothyris* and *Burmirhynchia* that indicate the Middle Jurassic age. The Longhai Formation comprises yellowish shale and siltstone, and yields a rich pelecypod fauna of late Middle Jurassic age in the lower part. The Middle Jurassic sequence in the Baoshan area is correlated with the Namyan Series in northern Shan state, Burma [10]. The disconformably overlying Longkan Formation is unfossiliferous and is tentatively assigned to the Upper Jurassic.

The Jurassic and middle Cretaceous rock formations were sampled from the Simao Depression, and the Middle Jurassic Mengga Formation was sampled from the Baoshan Block. The sampling details are given in the next section.

The climate of South and Southwest Yunnan is dominated by the monsoon from the Indian Ocean and is characterized by annually alternating rainy and dry seasons. Weathering thus presents a problem for sampling in this region, and the samples were therefore taken from stream beds and fresh road cuttings. A portable rock drill was used for sample extraction, and the cores were oriented with a magnetic compass. Five to six cores were usually collected from each site.

3. Treatment and analysis

The laboratory work was conducted at the University of Florida. Stepwise thermal demagnetization was used throughout the investigation using a TSD-1 Schonstedt thermal demagnetizer, and sample remanence was measured in an ScT two-axis cryogenic magnetometer. All the equipment is housed in a magnetically shielded room with a residual field of 50–250 nT. Orthogonal projection [17] was employed to examine the demagnetization behavior of the samples. Paleomagnetic directions and planes were calculated using principal component analysis (PCA) [18]. Site and formation means were computed using standard Fisherian [19] statistics, with the exception of two sites whose mean directions were determined using McFadden and McElhinny's method [20]. Sample directions with maximum angular deviation (MAD) or site mean directions with α_{95} values larger than 15° were rejected for the calculation of site or formation means except

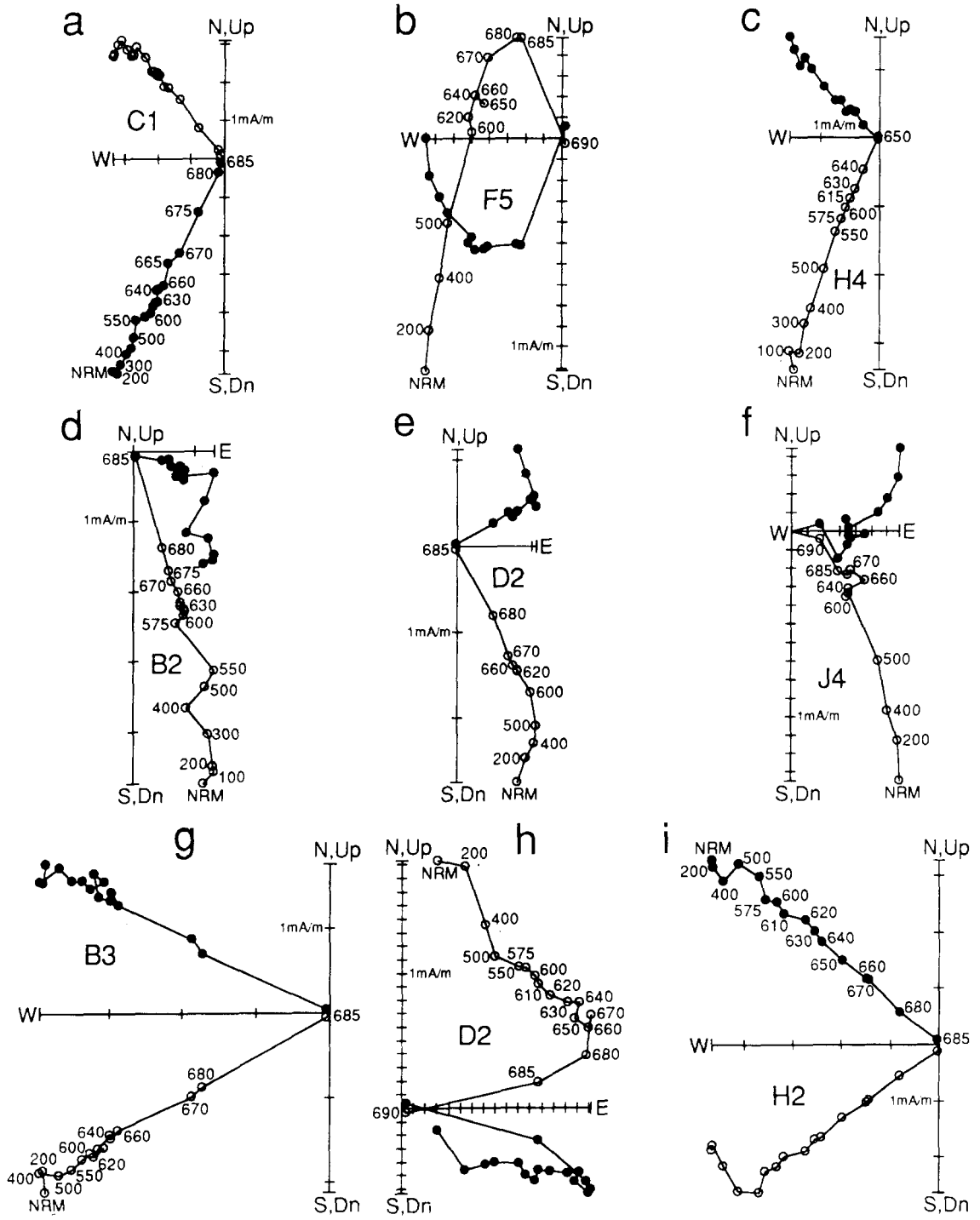


Fig. 2. Representative orthogonal plots of thermal demagnetization (in geographic coordinates) for the middle Cretaceous Nanxing Formation (a, b and c), and Mangang Formation from Jinggu (d, e and f) and Mengla (g, h and i). Closed symbols are plotted on horizontal plane, open symbols on the vertical plane. Thermal treatment levels are in degrees Celsius.

for the Middle Jurassic Mengga Formation from the Baoshan Block.

4. Paleomagnetic results

4.1 Middle Cretaceous

Middle Cretaceous samples were taken from the Mangang Formation and stratigraphically equivalent rocks from three localities. Most of them yielded good results.

Nanxing Formation, Xiaguan City (25.6°N, 100.2°E): In the vicinity of Xiaguan City, the stratigraphic equivalent of the middle Cretaceous Mangang Formation is the Nanxing Formation. Paleomagnetic samples from this formation were collected from two limbs of a small, gentle syncline outcropping on the southern shore of Lake Erhai, 3 km northeast of downtown Xiaguan. Lake Erhai is the result of subsidence due to the post-Miocene normal faulting on the Diancangshan segment of the Red River fault zone [21].

TABLE 1

Site and formation statistics for the middle Cretaceous formations

Site	Bedding	n/N	In Situ		Tilt Corrected				Pole Position	
			D(°)	I(°)	D(°)	I(°)	α_{95}	k_2	Lat.(°)	Long.(°)
The Nanxing Formation, Xiaguan (25.6°N, 100.2°)										
KDA	165/34	5/5	205.1	-47.7	159.0	-58.8	3.4	498.1	-67.6	233.8
B	165/34	4/4	204.5	-44.6	163.2	-56.7	7.6	149.1	-71.6	233.5
C	165/34	5/5	212.2	-27.8	189.8	-48.6	5.9	171.2	-80.5	343.5
D	165/34	6/6	222.7	-11.7	212.6	-39.1	13.3	26.4	-60.1	9.9
E*	307/36	3/5	256.1	74.8	14.7	64.2	29.0	19.2	66.5	126.4
F**	307/36	5/5	158.6	-70.4	194.7	-41.2	9.9	129.6	-76.5	15.4
G	307/36	4/4	299.6	71.1	5.4	52.2	4.0	542.2	81.4	132.1
H	307/36	5/5	311.1	67.8	3.3	47.2	5.0	233.2	86.0	146.2
I	303/41	5/5	308.5	68.3	3.0	42.7	5.2	220.4	87.2	206.6
J	303/41	5/5	341.5	69.9	13.9	34.6	5.6	190.3	75.6	214.5
Normal Sites:		4	315.0	67.0	6.7	44.3	9.6	93.5	84.0	184.9
Reversed Sites:		5	207.7	-41.7	187.1	-50.5	16.2	23.3	-81.6	326.5
[a]Formation Mean:		9/10	8.9	59.2	6.9	47.7	8.6		83.6	152.7
The Mangang Formation, Jinggu (23.4°N, 100.9°E)										
KMA	332/38	4/5	164.4	66.2	100.5	50.7	13.7	45.9	3.7	158.1
B	335/46	5/5	108.5	72.2	79.1	30.1	11.5	45.5	16.1	179.9
C*	330/52	5/5	184.7	55.3	106.2	49.6	17.4	20.3	-1.1	156.8
D	353/38	5/5	74.0	66.5	78.9	28.7	14.0	30.9	16.0	180.8
E	27/15	5/5	73.2	48.0	81.9	36.4	10.0	59.9	15.0	175.0
F	27/15	5/5	51.6	55.5	67.6	47.3	8.5	82.3	29.8	170.4
G	9/13	6/6	56.4	58.7	67.1	48.3	9.7	48.9	30.4	169.5
H	0/29	5/5	57.4	68.5	74.7	41.6	5.1	226.0	22.5	173.5
I	30/23	4/5	43.5	74.8	89.5	59.8	4.3	459.8	15.4	152.8
J*	332/32	5/5	353.5	50.7	17.8	32.2	54.2	3.0	72.3	207.2
[b]Formation Mean:		8/10	73.2	67.7	79.4	43.3	9.1		18.9	170.0
The Mangang Formation, Mengla (21.6°N, 101.4°E)										
KLA	354/119	6/6	277.5	27.5	69.9	32.0	4.2	261.2	24.5	181.5
B	354/119	5/5	294.0	24.9	52.8	28.9	11.0	49.5	39.8	189.0
C	346/122	5/5	274.7	5.7	47.3	48.3	7.2	115.0	46.9	171.1
D	348/125	6/6	*115.3	-29.7	224.7	-16.4	18.3	14.3	-45.0	21.5
			**111.7	-12.5	217.9	-32.8	7.1	125.2	-54.2	10.5
E	359/132	6/6	287.0	14.5	68.5	31.1	9.1	54.7	25.6	182.5
F	0/127	5/5	294.9	7.3	57.0	39.9	6.1	159.5	37.5	178.8
G	0/109	5/5	297.9	21.7	54.8	41.0	13.5	33.2	39.7	178.2
H	3/112	6/6	301.3	29.5	64.0	31.7	6.1	120.2	29.9	183.4
I	3/112	5/5	279.0	22.3	85.1	45.3	8.7	77.9	13.7	167.7
J	3/112	6/6	290.1	24.7	72.5	40.4	4.0	287.8	23.7	174.8
[c]Formation Mean:		10/10	288.7	19.3	60.8	37.8	7.6		33.7	179.3

Bedding = strike/dip with dip direction 90° clockwise from strike; n/N = number of samples or sites in calculation for the means/number of samples or sites measured; D and I = declination and inclination; α_{95} and A_{95} = radius of the circle of 95% confidence about the mean direction and pole, respectively; k_1 and k_2 = Fisherian precision parameters before and after tilt correction.

* Excluded from calculation of the formation mean.

** Computed using McFadden and McElhinny's 1988 method.

[a] $k_1 = 7.3$, $k_2 = 36.9$, $A_{95} = 10.0$.

[b] $k_1 = 21.6$, $k_2 = 38.3$, $A_{95} = 8.9$.

[c] $k_1 = 45.8$, $k_2 = 41.4$, $A_{95} = 8.2$.

Four sites (KDA–D) were drilled in the purplish red sandstone interbedded with conglomerate on the east limb, which dips southwest at 28–34°, and six sites (KDE–J) were taken in the dark purple siltstone on the west limb, which dips northeast at 36–41°. Forty-nine samples in total were obtained.

The intensity of the natural remanent magnetization (NRM) of the samples was mostly in the range 2.4–8.5 mA/m. The demagnetization behavior of the samples from eight sites is straightforward: after removal of a low temperature component at 300°C, a characteristic remanent magnetization (ChRM) component was isolated at higher temperatures (Figs. 2a and c). Sites A, B, C and D are of reverse polarity, and the ChRM components often unblocked in the temperature range 650–690°C (Fig. 2a). Samples from sites G, H, I and J, on the other hand, are of normal polarity, and the ChRM components unblocked in a slightly lower temperature range (Fig. 2c). In both cases the *in-situ* ChRM directions are different from that of the present-day Earth field. Samples from sites E and F were relatively weakly magnetized, with NRM intensity ranging from 1.0 to 2.9 mA/m. A reverse direction was consistently resolved from all the samples of site F at above 670°C but could not be adequately determined by the PCA technique (Fig. 2b). The sector-constrained great circle intersection method [20] was then applied to retrieve the mean direction of this site. Samples from site E failed to give internally consistent ChRM directions and were therefore excluded from the calculation of the formation mean (Table 1). After tilt correction, the mean direction for the Nanxing Formation based on the remaining nine sites is $D/I = 6.9°/47.7°$ ($\alpha_{95} = 8.6°$), which corresponds to a paleomagnetic pole at 83.6°N, 152.7°E ($A_{95} = 10.0°$), with a significant increase in Fisherian precision parameters ($k_2/k_1 = 5.05 > 3.37$, test value from [22]), so the fold test is positive at the 99% confidence level (Fig. 3a and Table 1). The angular distance between the mean directions of the normal and reverse groups after flipping one into the other is $7.2 \pm 25.8°$; thus the reversal test is also positive at the 95% confidence level (Table 1).

Mangang Formation, Jinggu county (23.4°N, 100.9°E): The sampling of this formation was carried out near a village called Yangdiaoyu along the highway from Puer to Jinggu. The local structure is basically a monocline, but the bedding attitude varies from dipping southeast at 23° to northeast at 52° owing to flexure. The NRM intensity of the samples ranged from 1.4 to 5.4 mA/m. Fifty samples from ten sites (KMA–J) were collected from purplish red muddy siltstone and silty mudstone. Site J did not give stable paleomagnetic directions at high temperatures due to the coarse-grained nature of the samples (Fig. 2f). Samples from other sites often became viscous above 600°C. Nonetheless, most samples from the remaining sites yielded well-defined high temperature components that are in general east-

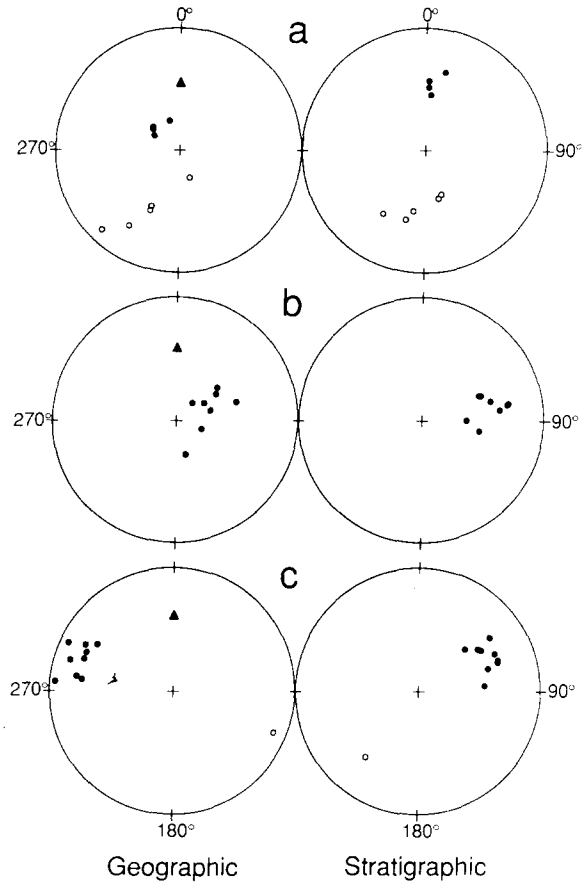


Fig. 3. Equal-area projections of site mean directions before and after tilt correction for the middle Cretaceous Nanxing Formation (a) and Mangang Formation from Jinggu (b) and Mengla (c). ● = directions plotted on the lower hemisphere; ○ = directions plotted on the upper hemisphere; ▲ = dipole field direction at the sampling localities.

erly with steep inclinations in geographic coordinates, and are different from the present-day Earth field direction at the sampling locality (Figs. 2d and e). Site C was excluded from the computation of the formation mean due to large within-site scatter ($\alpha_{95} = 17.4°$, see Table 1). The mean direction for this formation is based upon the eight acceptable sites. Tilt correction improves, al-

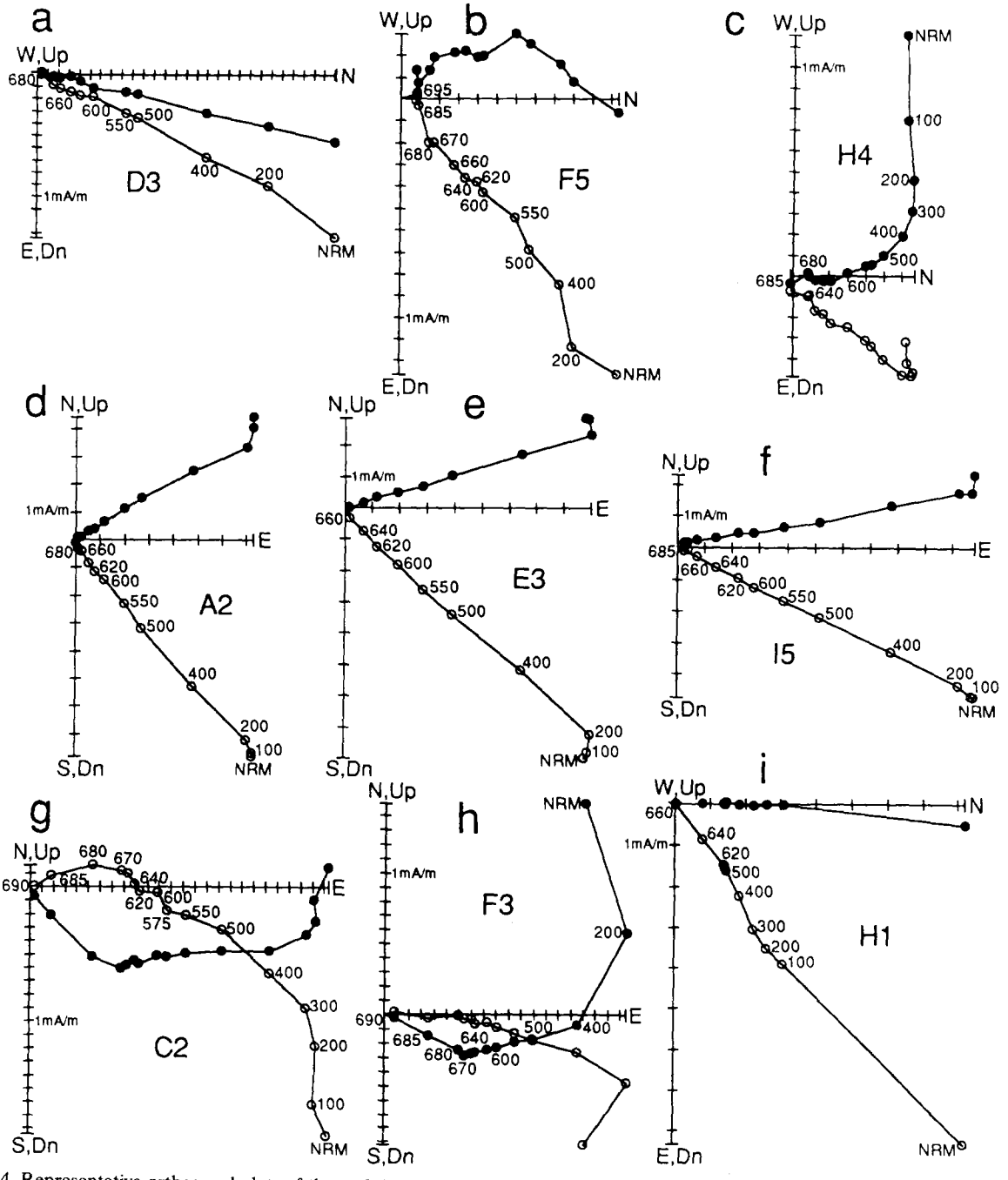


Fig. 4. Representative orthogonal plots of thermal demagnetzation for the Upper Jurassic Bazhulu Formation (a, b and c) and the Middle Jurassic Hepingxiang Formation (d, e and f) and Mengga Formation (g, h and i). Coordinates and symbols as in Fig. 2.

though not significantly, the grouping of the site mean directions (Fig. 3b and Table 1). The tilt-corrected mean direction for the Mangang Formation at this locality is $D/I = 79.4°/43.3°$, with $\alpha_{95} = 9.1°$, which yields a paleomagnetic pole at 18.9°N, 170.0°E with $A_{95} = 8.9°$.

Mangang Formation, Mengla county (21.6°N, 101.4°E): The Middle Cretaceous Mangang Formation was also sampled at Longlin, near the border between China and Laos. The sampled section is located on an overturned limb of a N–S trending syncline; the normal east limb was unfortunately inaccessible due to the dense tropical jungle. The rocks of the west limb dip easterly at 48–71°. Sampling was undertaken along the highway from Simao to Mengla. Fifty-five samples from ten sites (KLA–J) were collected from purplish red siltstone and sandstone. Most of the samples were strongly magnetized, with NRM intensity ranging from 2.0 to 9.1 mA/m. Thermal demagnetization resolved a ChRM from all the samples, with nine sites of normal polarity and one site (D) reversed. In the normal polarity group, the ChRM components were usually isolated above 400°C and unblocked in the temperature range 660–690°C (Figs. 2g and i). For site D, a low to medium temperature component persisted up to 660°C; the reverse component was consistently isolated from all the samples of this site and unblocked at above 670°C (Fig. 2h). The mean ChRM direction for this site calculated from the PCA-determined sample directions has a large α_{95} value (18.3°) and is of a much gentler inclination than that of the normal polarity group (see Table 1). The combined analysis of remagnetization circle and direct observation proposed by McFadden and McElhinny [20], however, yielded a mean direction for this site that is nearly antipodal to that of the normal polarity group, and the α_{95} value is much smaller (7.1°). The latter is believed to be a better estimate of the mean direction for this site, and was thus used to compute the formation mean. As can been seen from Fig. 3c, the site mean directions from this locality, whether before or after tilt correction, are distinctly different from that of the present-day Earth field. The tilt-corrected formation mean direction from this locality is $D/I = 60.8°/37.8°$ ($\alpha_{95} = 7.6°$), corresponding to a paleomagnetic pole at 33.7°N, 179.3°E ($A_{95} = 8.2°$).

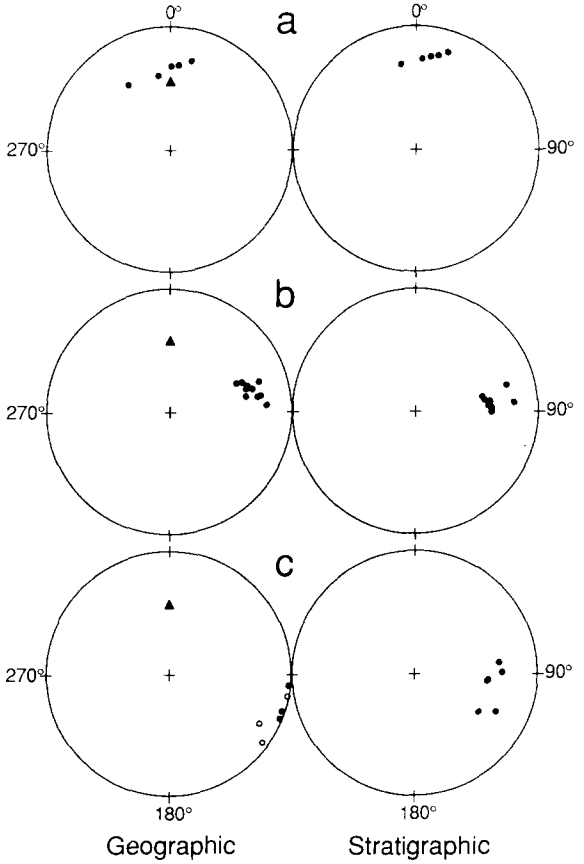


Fig. 5. Equal-area projections of site mean directions for the Upper Jurassic Bazhulu Formation (a), the Middle Jurassic Hepingxiang Formation (b) and the C component of the Mengga Formation (c). Symbols as in Fig. 3.

4.2 Jurassic

In the Simao Depression, the Jurassic rocks of the Lower, Middle and Upper Series were sampled from the area around Xiaguan City; also, Middle Jurassic rocks were sampled from Jinggu and Mengla counties. Unfortunately, only the Upper Jurassic samples from Weishan and the Middle Jurassic samples from Jinggu (Simao Depression) and Luxi (Baoshan Block) yielded useful results.

Upper Jurassic Bazhulu Formation, Weishan county (25.4°N, 100.2°E): The samples from this formation were collected from a section (Shuilazhuqing) located 20 km south of Xiaguan City; administratively this area belongs to Weishan county. The sampled structure is a monocline with dips of 14–22° to the northeast. Forty sam-

ples from eight sites (JUA–H) were taken from brick-red muddy siltstone and fine sandstone. These samples were relatively weakly magnetized, with most NRM intensity values in the range 0.5–3.2 mA/m. Samples from three sites (B, C and F) failed to give internally consistent directions (e.g., Fig. 4b) but the remaining five sites did yield usable results. During thermal demagnetization, most samples from these five sites exhibited more or less univectorial behavior (Fig. 4a); only a couple of samples displayed a prominent low temperature component, which was eliminated below 300°C (Fig. 4c). The *in-situ* directions of these samples are northerly but the inclinations are gentler than that of the present-day Earth field. The grouping of the site mean directions improves after correcting for bedding tilt (Fig. 5a and Table 2). The tilt-corrected mean direction for this formation derived from these five sites is $D/I = 7.3°/25.4°$ ($\alpha_{95} = 10.4°$), which corresponds to a paleomagnetic pole at 76.3°N, 250.0°E ($A_{95} = 10.4°$).

Middle Jurassic Hepingxiang Formation, Jinggu county (23.6°N, 100.5°E): This formation was sampled at the Wenzhaoying section located 25 km northeast of downtown Jinggu. The Hepingxiang Formation exposed at this section is more than 1000 m thick. The rocks range in dip from

TABLE 2

Site and formation statistics of the Jurassic formations

Site	Bedding	n/N	In situ D(°)	In situ I(°)	Tilt Corrected D(°)	Tilt Corrected I(°)	α_{95}	k_2	Pole Position Lat. (°)	Pole Position Long. (°)
The Upper Jurassic Bazhulu Formation, Weishan (25.4°N, 100.2°E)[d]										
JUA	327/22	4/6	351.2	39.1	3.9	27.7	4.2	485.7	78.7	260.6
D	339/14	4/4	13.4	26.9	18.1	18.5	10.5	77.5	66.6	229.8
E	327/33	4/5	326.7	37.7	349.6	31.0	9.3	98.1	77.0	330.4
G	340/17	3/5	0.9	32.3	9.4	25.0	8.3	222.6	74.9	242.7
H	340/17	5/5	6.1	31.3	13.7	22.8	15.1	26.5	71.3	234.0
[a]Formation Mean:		5/8	356.5	34.5	7.3	25.3	10.4		76.3	250.0
The Middle Jurassic Hepingxiang Formation, Jinggu (23.6°N, 100.5°E)										
JZA	15/20	5/5	67.2	40.6	74.4	24.0	2.2	1214.3	19.1	184.8
B	54/23	5/5	75.1	30.7	85.1	20.4	1.8	1743.0	8.6	182.7
C	90/26	5/5	71.5	33.9	90.6	38.1	2.5	922.7	7.9	170.5
D	93/12	5/5	73.8	35.2	82.5	38.3	5.1	222.8	15.0	173.2
E	93/12	5/5	78.8	36.5	87.9	38.5	3.1	611.4	10.3	171.2
F	101/14	5/5	68.0	36.8	78.5	43.4	1.7	2105.5	19.7	170.7
G	137/22	4/5	71.6	24.0	78.6	43.5	5.9	244.6	19.6	170.6
H	141/15	4/5	80.5	27.9	86.0	40.6	1.7	3038.4	12.5	170.4
I	163/17	6/6	79.9	25.1	81.4	42.0	3.6	351.0	16.8	170.9
J	163/17	5/5	86.0	21.5	88.4	38.0	5.1	225.3	9.8	171.4
[b]Formation Mean:		10/10	75.5	31.4	83.3	36.8	5.4		14.0	173.6
The Middle Jurassic Mengga Formation, Luxi (24.3°N, 98.4°E)[e]										
JMA	234/48	**C** 4/5	95.4	2.5	82.7	31.3	14.9	38.8	13.3	175.6
		B 4/5	84.0	14.0	63.8	31.5	7.2	165.1	30.4	182.2
B	234/48	**C** 4/5	108.3	2.9	94.9	39.5	19.3	23.7	4.9	166.0
		B 5/5	78.4	26.7	47.9	35.1	11.8	42.7	45.5	185.0
C	234/48	**C** 4/5	125.9	-6.5	120.7	38.8	22.3	18.0	-16.2	154.6
		B 5/5	83.6	16.7	61.1	32.9	14.3	29.5	33.1	182.1
D	240/52	**C** 5/5	118.2	-16.8	115.3	27.6	22.5	12.5	-15.8	163.8
		B 5/5	86.1	11.1	67.4	27.3	11.3	47.0	26.3	183.6
E	240/52	**C** 5/5	100.6	-2.0	89.4	29.4	14.6	28.4	6.9	174.2
		B 5/5	93.1	18.7	64.7	37.2	19.4	16.6	30.7	177.7
F	240/52	**C** 5/5	111.9	2.4	96.2	40.2	6.4	144.3	4.0	165.0
		B 5/5	81.9	16.5	59.8	27.2	7.5	104.7	33.1	186.6
[c]Formation Mean:		**C** 6/6	110.0	-2.9	99.7	35.2	11.3		-0.5	166.6
		B 6/6	84.6	17.3	60.9	32.0	5.9		33.2	182.8

C and B = components C and B of the remanence of the Mengga Formation. Other expanations as in Table 1.

[a] $k_1 = 26.9$, $k_2 = 54.9$, $A_{95} = 10.4°$.

[b] $k_1 = 98.2$, $k_2 = 81.5$, $A_{95} = 4.2°$.

[c] For component C, $k_1 = 35.8$, $k_2 = 35.9$, $A_{95} = 12.2°$, and for component B, $k_1 = 132.2$, $k_2 = 132.3$, $A_{95} = 5.8°$.

[d] Sites B, C and F were omitted due to large within-site inconsistency.

[e] Sites G–K from the Datianyakou section were omitted due to severe present-day overprinting.

southeastward to southwestward, at 12–26°. Fifty-one samples from ten sites (JZA–J) were collected from purplish red, locally yellowish, calcareous siltstone and mudstone of the Lower Member of the formation. These samples were very strongly magnetized, with NRM intensity mostly in the range 11.4–24.3 mA/m. Thermal demagnetization isolated a ChRM component above 200°C from nearly all the samples after a low temperature component was removed (Figs. 4d, e and f). The ChRM directions are easterly with moderate downward inclinations in geographic coordinates, and are apparently different from that of the present-day Earth field (Fig. 5b). The tilt-corrected mean direction for this formation is $D/I = 83.3°/36.8°$ ($\alpha_{95} = 5.4°$), which yields a paleomagnetic pole at 14.0°N, 173.6°E ($A_{95} = 4.2°$) (Table 2).

Middle Jurassic Mengga Formation, Luxi City (24.3°N, 98.4°E): This formation was sampled at two sections, the Mengga–Longhai type section 25 km southwest of downtown Luxi (24.3°N, 98.4°E) and the Datianyakou section 35 km northeast of Longling county (24.8°N, 98.9°E). The structure at both sections is a monocline with moderate dip, with a northwest dip in the first section and a northeast dip in the second section. The samples are purplish red fine sandstone or calcareous siltstone and come from the lower part of the formation. Thirty samples from six sites (JMA–F) were collected from the Mengga–Longhai section and 25 samples from five sites (JMG–K) were taken from the Datianyakou section. It was noted that the rocks sampled at the second locality were weathered. Thermal demagnetization revealed three components from most of the samples from the Mengga–Longhai section: a low temperature component (A) is directed in the present-day Earth field and was eliminated below 200°C; an intermediate temperature component (B), which is easterly with gentle inclinations in geographic coordinates, was seen in the temperature range 400–660°C; and a ChRM component (C), which is easterly or east-southeasterly with very gentle inclinations in geographic coordinates, was isolated above 670°C (Figs. 4g and h). The NRM intensity of the samples from the Datianyakou section was dramatically decreased after heating to only 100°C, indicating the existence of a magnetic mineral of low Neel temperature, possibly goethite. The *in-situ* paleomagnetic direction was still very close to that of the present-day Earth field even when the samples were heated to 660°C (Fig. 4i), probably as a consequence of deep weathering. Therefore, these five sites were not included in the calcula-

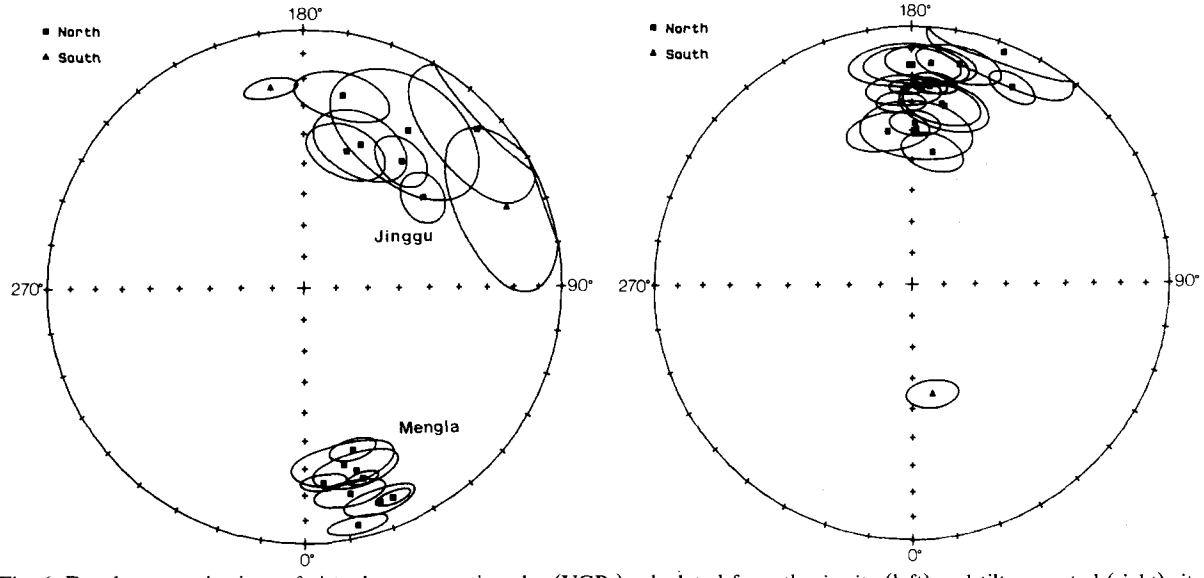


Fig. 6. Equal-area projections of virtual geomagnetic poles (VGPs) calculated from the *in-situ* (left) and tilt-corrected (right) site directions of the middle Cretaceous Mangang Formation from Jinggu and Mengga. Squares and triangles represent the north and south poles respectively.

tion of the mean direction for the Mengga Formation. Based on the six sites from the Mengga–Longhai section, the tilt-corrected mean direction of the B component is $D/I = 60.9°/32.0°$ ($\alpha_{95} = 5.9°$), and that of the C component is $D/I = 99.7°/35.2°$ ($\alpha_{95} = 11.3°$). The corresponding paleomagnetic poles are 33.2°N, 182.8°E ($A_{95} = 5.8°$) and 0.5°S, 166.6°E ($A_{95} = 12.2°$), respectively (Fig. 5c and Table 2).

5. Stability of magnetization

In recent years paleomagnetists have become increasingly more aware of the problem of remagnetization in paleomagnetic study [e.g., 5,23, 24], and have attached more importance than ever to various stability tests of magnetization. The ChRM isolated from the Nanxing Formation from Xiaguan passes the fold test and is thus pre-folding or pre-Eocene in age. However, half of the acceptable sites are reversely magnetized. This is at odds with the mapped middle Cretaceous age of the formation (Aptian to Albian) which is supposed to be within the Cretaceous Long Normal Superchron (CLNS) [25]. It is probable that the ChRM of the Nanxing Formation was acquired during latest Cretaceous times. The *in-situ* ChRM directions of the Mangang Formation from Jinggu and Mengla deviate strikingly from the present field, suggesting that they are not a recent overprint. The sampled rocks at these two localities have distinct bedding attitudes; it is therefore possible to perform a regional tilt test on the ChRM. Seeing as the two localities are 200 km apart, it is better to compare the site virtual geomagnetic poles (VGPs) instead of mean directions. It becomes apparent from Fig. 6 that the site VGPs calculated from the *in-situ* directions of the two localities form two well-separated groups, whereas the VGPs com-

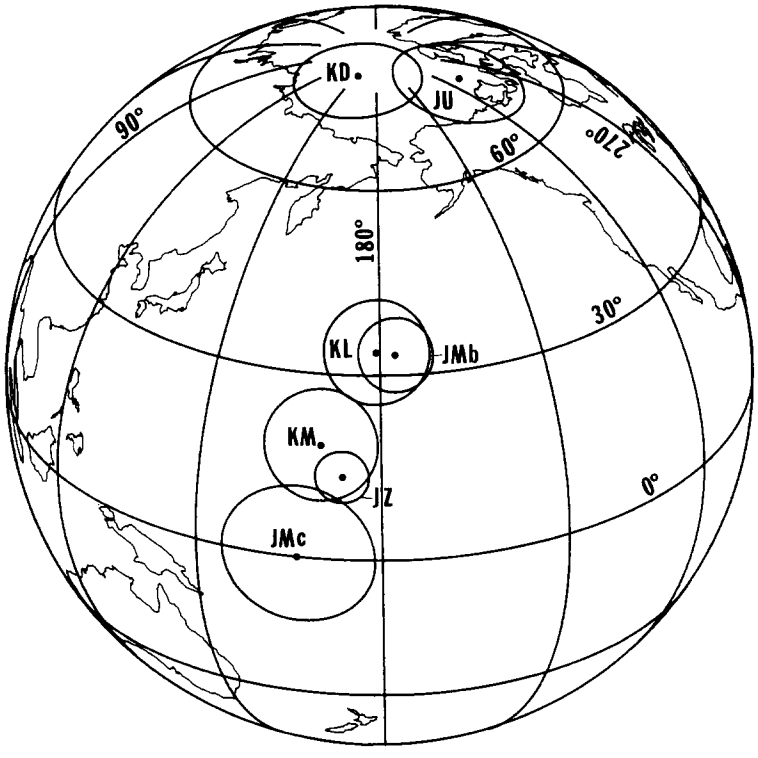


Fig. 7. Paleomagnetic poles and associated 95% confidence circles determined from the Cretaceous and Jurassic formations of this study. *KD* = Middle Cretaceous Nanxing Formation; *KM* = middle Cretaceous Mangang Formation from Jinggu; *KL* = middle Cretaceous Mangang Formation from Mengla; *JU* = Upper Jurassic Bazhulu Formation; *JZ* = Middle Jurassic Hepingxiang Formation; *JMb* = B component of the Middle Jurassic Mengga Formation; *JMc* = C component of the Mengga Formation. Equal-area projection.

TABLE 3

Paleomagnetic rotation and latitude difference calculated for selected localities in Tibet and surrounding areas

Nr.	Slat.°N/Slong.°E	Rock Age	Eurasia Pole(Ma)	D_{obs} °	D_{exp} °	Rotation °	Significant?	P_{obs} °	P_{exp} °	Lat. Difference °	Significant?	Reference
Tarim-Dzungar												
1*	44.2/86.0	K_u-E	80	12.5±11.0	16.1±4.3	-3.6±11.8	no	32.0±6.4	37.6±3.4	5.6±7.2	no	[33]
2	44.2/86.0	J_u-K_l	120	12.7±8.3	15.1±5.0	-2.4±9.7	no	29.6±4.8	34.4±4.1	4.8±6.3	no	[33]
3*	39.0/75.6	K	100	11.2±6.1	13.3±5.9	-2.1±8.5	no	21.2±3.3	31.3±5.0	10.1±6.0	yes	[50]
4*	41.6/83.5	K_u	80	16.0±11.6	15.2±4.1	0.8±12.3	no	22.0±6.4	34.6±3.4	12.6±7.2	yes	[32]
5	41.8/82.0	J_u-K_l	120	22.0±12.1	13.8±4.8	8.2±13.0	no	24.2±6.8	31.4±4.1	7.2±7.9	no	[32]
Pamirs												
6*	38.6/68.9	K_l	70	350.5±7.0	12.4±4.8	-21.9±8.5	yes	35.9±4.1	30.9±4.1	-5.0±5.8	no	[29]
7*	39.5/73.3	K_l	80	92.0±9.2	13.0±4.0	79.0±4.0	yes	34.5±5.3	30.7±3.4	-3.8±6.3	no	[28]
8	39.6/72.3	K_l	80	0.0±11.6	12.9±4.0	-12.9±12.3	yes	34.5±6.7	30.6±3.4	-3.9±7.5	no	[28]
9*	38.8/70.9	K_l	80	7.0±4.2	12.5±3.9	-5.5±5.7	no	25.8±2.4	29.6±3.4	3.8±4.2	no	[28]
10*	37.9/70.1	K_l	80	316.0±7.1	12.2±3.9	-56.2±8.1	yes	24.2±4.0	28.6±3.4	4.4±5.2	no	[28]
Tibet												
11*	33.8/80.4	K	100	6.3±5.9	13.3±5.6	-7.0±8.1	no	10.6±3.0	27.1±5.0	16.5±5.8	yes	[51]
12	32.0/91.5	K_m	100	338.7±9.9	14.5±5.6	-35.8±11.4	yes	13.4±5.2	27.6±5.0	14.2±7.2	yes	[44]
13*	30.0/91.0	K_m	100	354.3±9.0	14.2±5.5	-19.9±10.5	yes	11.8±4.7	25.6±5.0	13.8±6.9	yes	[44]
14	29.9/91.2	K_m	100	357.0±6.9	14.2±5.5	-17.2±8.8	yes	7.6±3.5	25.5±5.0	17.9±6.1	yes	[31]
15	29.7/98.5	K_l	110	57.7±17.0	17.4±5.6	40.3±17.8	yes	31.0±9.8	23.5±5.1	-7.5±11.0	no	[46]
16*	29.7/98.7	K_m	80	39.7±13.7	15.2±3.8	24.5±14.2	yes	32.2±7.9	26.4±3.4	-5.8±8.6	no	[46]
Indochina												
17*	25.6/100.2	K_m	70	6.9±12.8	14.1±4.5	-7.2±13.6	no	28.8±7.3	24.4±4.1	-4.4±8.4	no	This study
18*	23.4/100.9	K_m	80	79.4±12.5	14.6±3.6	64.8±13.0	yes	25.2±7.0	20.8±3.4	-4.4±7.8	no	This study
19*	21.6/101.4	K_m	80	60.8±9.6	14.5±3.6	46.3±10.3	yes	21.2±5.3	19.2±3.4	-2.0±6.3	no	This study
20*	16.5/103.0	K_m	100	28.1±3.2	13.7±5.2	14.4±6.1	yes	23.1±1.8	15.1±5.0	-8.0±5.3	yes	[34]
21*	1.5/110.0	J_u-K_u	120	270.8±9.7	14.9±4.1	-104.1±10.5	yes	-6.6±4.9	-1.4±4.1	5.2±6.4	no	[7]**
22	25.4/100.2	J_u	140	7.3±11.5	20.3±12.0	-13.0±16.6	no	13.3±6.0	29.4±10.4	16.1±12.0	yes	This study
23	23.6/100.5	J_m	160	83.3±6.7	20.9±7.3	62.4±9.9	yes	20.5±3.7	34.5±6.0	14.0±7.0	yes	This study
24	24.3/98.4	J_m(B)	80	60.9±7.0	14.6±3.6	46.3±7.9	yes	17.4±3.7	21.1±3.4	3.7±5.0	no	This study
25	24.3/98.4	J_m(C)	160	99.7±13.8	21.4±7.3	78.3±15.6	yes	19.4±7.5	34.4±6.0	15.0±9.6	yes	This study
South China												
26*	30.0/102.9	J_u-E	110	12.7±4.0	17.9±5.6	-5.2±6.9	no	18.8±2.2	24.9±5.1	6.1±5.6	yes	[52]
27*	26.5/102.4	$K_{u\text{-}m}$	100	8.1±8.5	14.6±5.5	-6.5±10.1	no	21.9±4.7	24.7±5.0	2.8±6.9	no	[53]
28	26.8/102.5	K_l	110	22.1±4.8	17.5±5.5	4.6±5.0	no	20.7±2.6	21.7±5.1	1.0±5.7	no	[53]
29*	32.0/119.0	K_u	80	13.7±14.3	16.3±4.1	-2.6±14.9	no	38.7±8.3	33.4±3.4	-5.3±9.0	no	[54]
30*	29.7/120.3	K_l	120	13.6±9.0	17.3±4.7	-3.7±10.2	no	28.4±5.1	28.4±4.1	0.0±6.5	no	[55]
31*	22.2/114.1	J_u-K_l	80	11.5±13.0	14.9±3.7	-3.4±13.5	no	28.9±7.4	22.8±3.4	-6.1±8.1	no	[56]
32*	35.9/128.5	K_l	120	29.5±9.2	18.8±5.1	10.7±10.5	yes	37.6±5.3	36.4±4.1	-1.2±6.7	no	[35]
33*	25.0/101.5	$K_{m\text{-}l}$	110	44.6±14.2	17.1±5.4	27.5±15.2	yes	23.7±7.9	19.7±5.1	-4.0±9.4	no	[47]
North China-Mongolia												
34*	45.4/107.6	J_u-K_l	120	7.7±8.7	19.5±5.4	-11.8±10.2	yes	42.2±4.9	40.1±4.1	-2.1±6.4	no	[36]
35*	42.0/119.2	K_l	120	6.8±9.0	19.9±5.3	-13.1±10.4	yes	39.2±5.2	39.8±4.1	0.6±6.6	no	[37]
36*	35.9/119.4	K_l	120	26.1±21.6	18.4±5.0	7.7±22.2	no	36.2±12.5	34.1±4.1	-2.1±13.2	no	[55]
37*	35.0/108.0	K_l	120	16.4±10.7	17.2±4.8	-0.8±11.7	no	31.0±6.1	30.4±4.1	-0.6±7.3	no	[57]

Nr. = Numbers (with an asterisk) denoting sampling localities in Fig. 8; *Slat. / Slong.* = latitude and longitude of sampling localities; D_{obs} and P_{obs} = oberved declination and paleolatitude; D_{exp} and P_{exp} = predicted declination and paleolatitude from Eurasia APWP [30]. Positive rotation is clockwise, and positive latitude difference indicates northward translation. Estimation of uncertainties follows Beck [58] and Achache et al. [44].

** Recalculated from three sites from the Pedawan Formation and two sites from the Bau Limestone Formation.

puted from the tilt-corrected site mean directions converge. The Fisherian precision parameter k increases 13-fold from 2.1 to 27.4 after tilt correction, which is much greater than the test value (2.26) tabulated by McElhinny [22] for a fold test to be positive at the 99% confidence level. How-

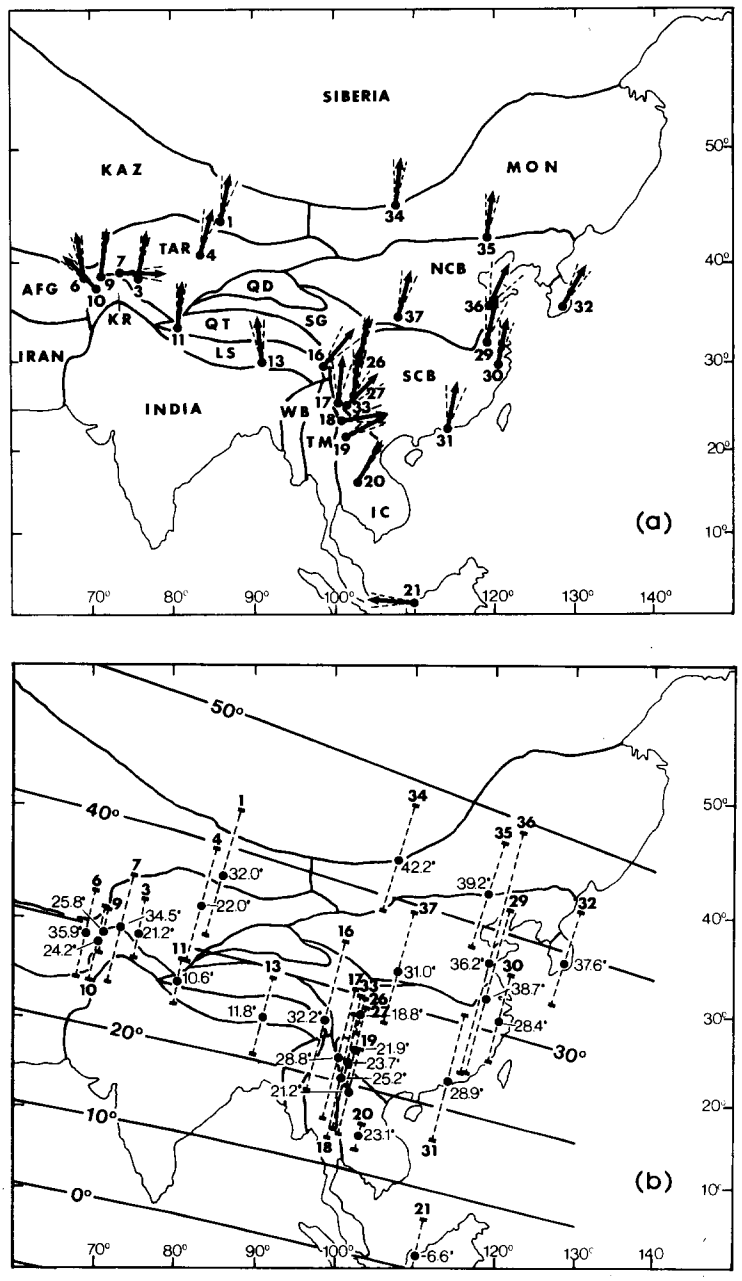


Fig. 8. Observed Cretaceous declinations (a) and paleolatitudes (b) from Tibet and surrounding regions. Bold-face numbers denoting the sampled locations are those in Table 3. Uncertainties associated with the observations are indicated with broken lines. Paleolatitudinal circles predicted from the 80 Ma Eurasian pole [30] are also indicated in (b). Base map is after Lin and Watts [31]. Abbreviations for the major terranes: *AFG* = Afghanistan; *IC* = Indochina; *KAZ* = Kazakhstan; *KR* = Karakoram; *LS* = Lhasa; *MON* = Mongolia; *NCB* = North China Block; *QD* = Qaidam; *QT* = Qiangtang; *SCB* = South China Block; *SG* = Songpan-Garze (Complex); *TAR* = Tarim; *TM* = Shan–Thai–Malay; *WB* = West Burma. Mercator projection.

ever, the value of the statistic proposed by McFadden and Jones [26] for determining significance of the fold test is 0.4464 ($R_a = 7.8224$, $R_b = 9.7496$, $R = 17.3799$, $N = 18$), which is greater than the threshold (0.2059) at the 95% significance. This means that the hypothesis of a common mean pole for the VGPs from the two sections can be rejected. McFadden and Jones [26] demonstrate that the statistical test suggested by McElhinny [22] is inappropriate. In fact, the paleomagnetic poles determined from the Mangang Formation at the two sections are tangential to each other (Fig. 7), with an angular distance of $17.0 \pm 17.1°$. There are two factors which might contribute to this discordance. First, the site mean directions and corresponding VGPs from Jinggu have larger α_{95} (A_{95}) values and are more scattered due to the viscous magnetization obtained at high demagnetization temperatures (Fig. 6 and Table 1). Second, the two sampled sections are not located on different limbs of a single fold, and there might be some relative rotation between them. Therefore, the ChRM resolved from the Mangang Formation is still most likely prefolding in age. Seeing as a reverse site occurs in the collection from Mengla, the acquisition of the ChRM may be Late Cretaceous in age.

There is no fold test available for the ChRMs isolated from the Middle Jurassic Hepingxiang Formation from Jinggu and the Mengga Formation from Luxi. The good internal consistency and striking deviation from the present field of the ChRM of the Hepingxiang Formation suggest its non-recent origin. The same can be said for the B and C components resolved from the Mengga Formation. More interestingly, the paleopole determined from the B component is coincident with that of the middle Cretaceous Mangang Formation and the paleopole derived from the C component overlaps that of the Middle Jurassic Hepingxiang Formation (Fig. 7). This indicates that the B component of the Mengga Formation might be a Cretaceous overprint coeval with the ChRM of the Mangang Formation, and that the C component of the Mengga Formation and the ChRM of the Hepingxiang Formation might have been acquired during Jurassic times. Finally, the mean direction for the Upper Jurassic Bazhulu Formation is based upon only five sites from a monoclinal section, and needs to be confirmed by future study. The fact that the inclination is gentler than that of the present-day field in geographic coordinates and the grouping of the site mean directions is improved after tilt correction, however, suggests that the ChRM of the Bazhulu Formation is probably not a recent overprint.

6. Tectonic implications

The quality and quantity of paleomagnetic data from Asia has increased dramatically in the last decade. Cretaceous data are now particularly numerous, covering nearly all the terranes that constitute Tibet and the surrounding regions. This allows us to compare the results from South and Southwest Yunnan with those from Tibet and its neighboring areas. These data are summarized in Table 3 and presented in Fig. 8 in terms of paleomagnetic rotation and latitude difference. Because our attention is focused on the effect of the India–Asia collision on the deformation of the Asian continent, results from the area south of the Indus–Zangbo Suture [e.g., 27] are not included. The selected data were obtained by employing magnetic cleaning techniques (primarily thermal demagnetization) and were calculated on site means (with the exception of the mean paleomagnetic directions from the Pamir–Punjab Syntaxis, which were calculated at the sample level [28,29]). Early acquisition of the magnetization in most studies was established with a positive fold test and dual polarity. A difficulty encountered in computing the predicted declination and paleolatitude is which specific Eurasian pole should be used for comparison. The choice is in essence what age should be assigned to the magnetization resolved from the individual collections. We used the Eurasian APWP proposed by Besse and Courtillot [30]. An important factor we took into account is the CLNS, i.e. if dual polarity is present either an Early or Late Cretaceous age was assigned, depending on the age of the rocks in question. This uncertainty should not change the general picture very much if the Cretaceous hairpin turn is a real feature of the Eurasian APWP [30].

The basic observations from Fig. 8 and Table 3 can be summarized as follows: First, the region directly in front of the Indian subcontinent, in-

cluding Tibet and Tarim, shows the most substantial northward translation after the Cretaceous but little rotation, except the Lhasa area which has obviously been rotated counterclockwise. Because several studies by different authors have obtained similar results, the large northward translation of this region appears to be beyond doubt. The paleomagnetically determined maximum translation for Tibet is 1970 ± 670 km [31], and that for Tarim is 1380 ± 790 km [32]. In Li et als'. study [32], however, the high temperature directions were not very stable and the calculated mean direction of the formation was based on the low temperature directions. We therefore consider that more than 1000 km of northward translation for Tarim is an overestimate, and several hundred kilometers is probably more realistic [33]. Second, the rotation of the sampled sites in the Pamir Syntaxis varies from unrotated to significantly rotated, and the general pattern seems to be that the sites on the western side of the syntaxis are rotated counterclockwise whereas those on the eastern side are rotated clockwise. None of the latitudinal variation is statistically significant. Paleomagnetic data from the northwestern Himalaya of India shows a similar rotation pattern [27]. Third, in a south-trending elongated zone east of the Yunnan Syntaxis, including eastern Qiangtang, central Yunnan, Indochina and northwestern Borneo, most of the sampled sites are rotated clockwise significantly, with the exception of Xiaguan, which is virtually unrotated, and northwestern Borneo, which is rotated counterclockwise by 104°. The Jurassic paleomagnetic results from South and Southwest Yunnan reported in this study show the same rotation pattern as the Cretaceous. Latitude differences seem to indicate that the area has been translated southward, but the amount of translation in most cases is well within the paleomagnetic uncertainties. The newly obtained Cretaceous results from northern Thailand constitute an exception and indicate 880 ± 580 km of southward translation [34]. Lastly, South and North China (including Mongolia) do not show any unambiguous rotation. Although results from a locality in South Korea [35] and two localities in Mongolia and North China [36,37] indicate some rotation, the amount is comparable to the paleomagnetic uncertainties. More data are needed to establish whether this is due to local rotation or whether it is of regional significance. No significant latitudinal variation has been detected for these two regions, with the exception of the Sichuan basin, which shows a small amount of northward translation that probably reflects convergence along the Qingling Suture and Longmenshan fold–thrust belt [38].

There is a consensus that deformation of the Cretaceous and younger rocks in Tibet and surrounding regions is intimately related to the collision of India with Asia and its subsequent northward indentation [1–3,39,40]. The large northward translation of the sampled sites in Tibet proper and Tarim reflects substantial crustal shortening of the region in the north–south direction. Two major mechanisms have been proposed to account for such a large amount of crustal shortening—crustal thickening through intracontinental thrusting/folding and/or vertical stretching [39–41] and lateral extrusion by strike-slip faulting [1–3]. The relative importance of these processes is still debated. The lack of rotation in this region may indicate either that the pre-collision leading edge of India was relatively smooth, as suggested by some authors [42,43], or that the presently existing crust of Tibet and Tarim has not been substantially moved in the east–west direction, or both.

The west-counterclockwise versus east-clockwise rotation pattern in the Pamir Syntaxis can be interpreted as an effect of oroclinal bending caused by the indentation of the northwestern tip of India. Similarly, the counterclockwise rotation of the Lhasa area [31,44] in concert with the clockwise rotation of the Markam area in the eastern Qiangtang Terrane [45,46] may be regarded as a similar effect caused by the northeastern tip of India. The lack of significant latitudinal change in the syntaxial areas may indicate that the crust there has stayed at the latitudes that were more or less expected from the Eurasian APWP or has been slightly extruded southward. If this were true, the much lower paleolatitude observed for the central part of Tibet than expected from stable Eurasia would suggest that the crust of central Tibet has been substantially shortened and thickened but not significantly moved out of the path of impinging India.

Cretaceous paleomagnetic data recently re-

ported from the Central Yunnan basin indicate that large clockwise rotation is not confined to the area south of the Red River fault [47]. Furthermore, it now becomes apparent that differential rotations have taken place within Sundaland (Table 3). For instance, relative to stable Eurasia the Jinggu and Mengla area of South Yunnan has been rotated clockwise by 46–65°, northern Thailand has been rotated by only 14° clockwise, and northwestern Borneo has been rotated counterclockwise by up to 104°. These observations are incompatible with the hypothesis of extrusion tectonics, which originally envisaged that Sundaland as a whole has been rotated by 40° since the India–Asia collision [2,3]. In addition, at face value, the Cretaceous paleolatitudes observed from Xiaguan, Jinggu and Mengla seem to indicate that South Yunnan has been systematically extruded southward by 220–480 km (about 310–680 km southeastward) (Fig. 8b and Table 3). As we have pointed out, however, the calculated translations are within the paleomagnetic uncertainties. Although the Cretaceous paleomagnetic results from the Khorat basin of northern Thailand [Yang and Besse, this issue] suggest 880 ± 580 km southward (about 1240 ± 820 km southeastward) translation with respect to stable Eurasia (Table 3), the significance of the latitudinal difference in that study is primarily derived from the exceptionally small α_{95} value (2.4°) associated with the mean paleomagnetic direction. We thus feel that 1000 km or more is probably an overestimate for the displacement along the Red River fault during its left-lateral motion phase, and the actual amount is probably less than 500 km. In a recent paper, Harrison et al. [48] reviewed the comtemporary tectonic models proposed to explain the thickening and uplift of the Tibetan Plateau. We agree with them in that none of these models can completely explain the present disposition of the crust in southern Asia. It seems to us that the crust in the syntaxial area has been largely squeezed out by the protruding tips of India but the lateral extrusion may not be as large as the proponents of extrusion tectonics originally envisaged. The clockwise rotation of Sundaland is more complex than the plasticine experiment suggested [2,49]. The crust in central Tibet and Tarim is basically shortened and thickened, with little lateral extrusion. Undoubtedly, much more work needs to be done before the role played by each mechanism can be better understood. From our perspective, future paleomagnetic investigations should be intensified in the zones bounding the Indian indentor and be expanded into the Cenozoic sedimentary rocks.

7. Conclusions

(1) Paleomagnetic directions from the middle Cretaceous formations of Southern Yunnan indicate that the Xiaguan area, which is situated on the Red River fault, has not been rotated with respect to stable Eurasia since at least Late Cretaceous time, and that the Jinggu–Mengla area (200–400 km south) has been rotated by 46–65° in a clockwise direction. The Jurassic results from the region show a similar rotation pattern. The differences between the observed and predicted Cretaceous paleolatitudes seem to show some southward translation of the region, but the amount of translation is not large enough for paleomagnetic methods to ascertain.

(2) Comparison with the Cretaceous data from Central Yunnan, northern Thailand and northwestern Borneo suggests that the Red River fault is not a demarcation between unrotated and significantly rotated regions, and that Sundaland was not deformed as a coherent block in response to the collision of India with Asia and its subsequent northward penetration.

(3) The available Cretaceous paleomagnetic data from East Asia indicate substantial northward translation but little rotation for central Tibet and Tarim, and large differing rotations but not much paleomagnetically detectable latitudinal variation for the syntaxial areas. The existing paleomagnetic data also reveal neither significant rotation nor latitudinal change for North and South China since the Cretaceous. We interpret these observations as indicative of substantial crustal shortening and thickening having taken place in central Tibet and a certain amount of crustal extrusion in the syntaxial areas.

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